

BattingEdge Biomechanical Diagnostic Report

Phase: Post-V8p Inference Audit

Date: November 25, 2025

Author: Anoshia (with Copilot)

Project Objective

The BattingEdge project aims to classify cricket shots — **Drive, Pull, Cut, Sweep** — from video clips using pose estimation and machine learning. The goal is to build a robust, reproducible system that works even on short, noisy clips and generalizes across batters and environments.

Original Pipeline

1. **Pose Extraction:** MediaPipe Pose used to extract 33 body landmarks per frame.
2. **Feature Engineering V8p:**
 - 99 pose features (x, y, z for each landmark)
 - 4 biomechanical features:
 - Wrist velocity
 - Knee angle
 - Bat angle (proxy)
 - Stance width
 - Resampled to 50 frames × 103 features
3. **Model Training V8p:**
 - Bi-LSTM classifier trained on balanced dataset
 - Achieved 78% test accuracy
4. **Inference Pipeline:**
 - Extracts pose + biomech features from new clips
 - Applies scaler and encoder
 - Predicts shot class with confidence

Problem Statement

Despite good test accuracy, the model **frequently misclassified Drive vs Pull** during inference. This raised concerns about:

- Feature overlap between shot types
- Pose ambiguity in short clips
- Inadequate biomechanical separation
- Overfitting to training posture or viewpoint

Diagnostic Goal

To **audit biomechanical features across all shot classes** and identify:

- Which features are reliable separators
- Where confusion zones exist
- How to improve feature logic for better inference

Methodology

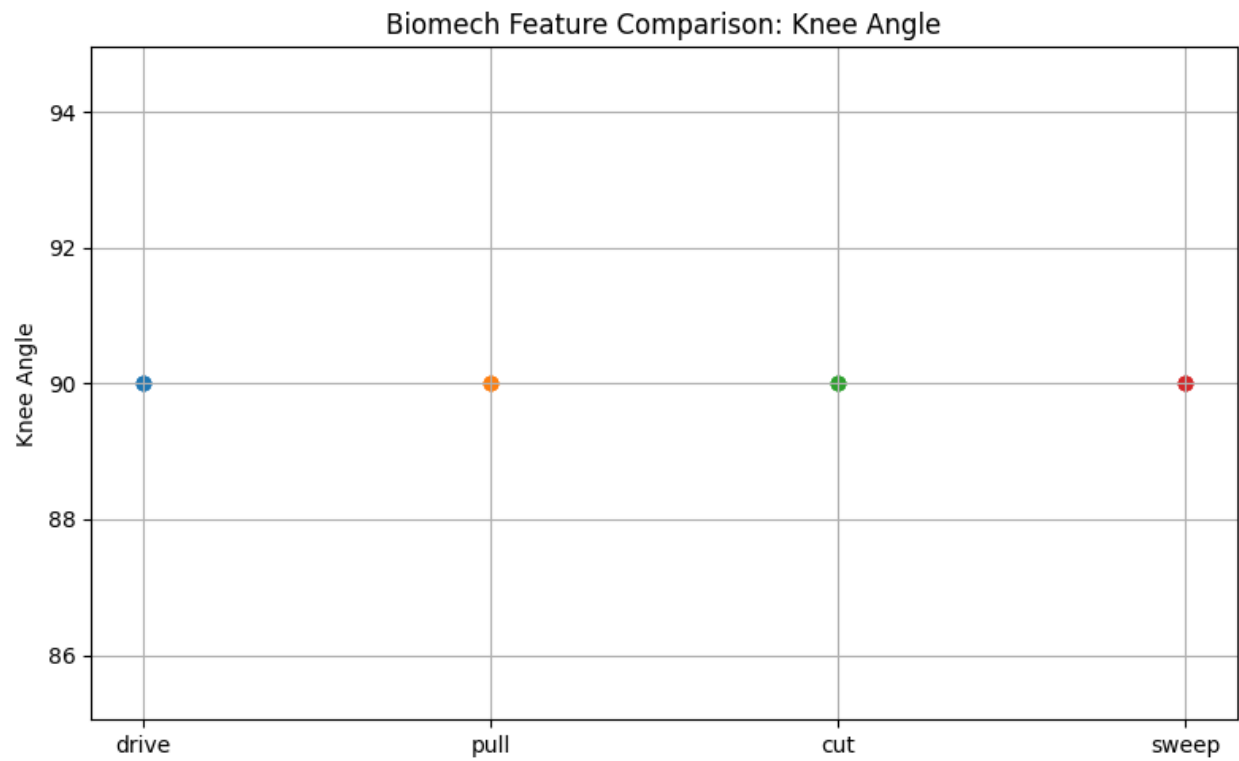
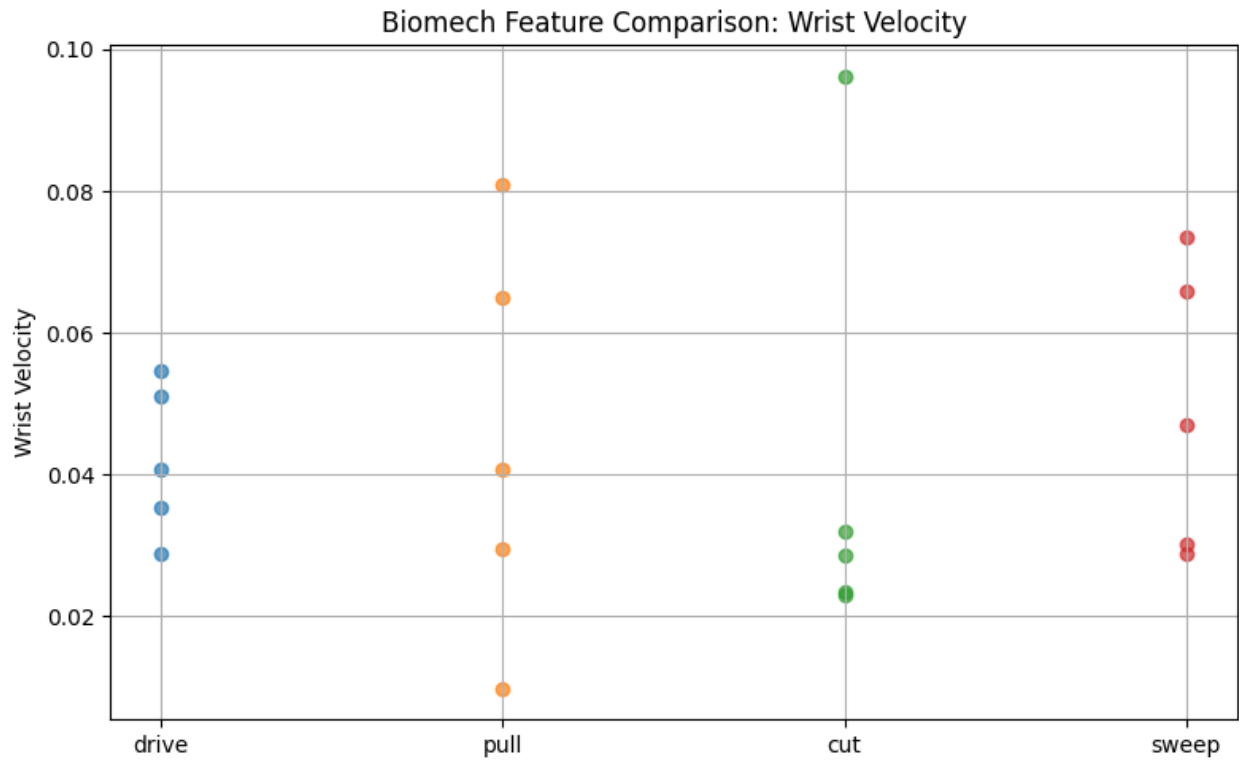
1. Selected 5 test videos per class from dataset_v7_clean/test
 2. Extracted frame-wise pose landmarks using MediaPipe
 3. Computed 4 biomechanical features:
 - Wrist velocity (frame-to-frame wrist motion)
 - Knee angle (hip–knee–ankle)
 - Bat angle (shoulder–wrist proxy)
 - Stance width (ankle-to-ankle distance)
 4. Averaged features per video
 5. Plotted scatter comparisons across classes
-

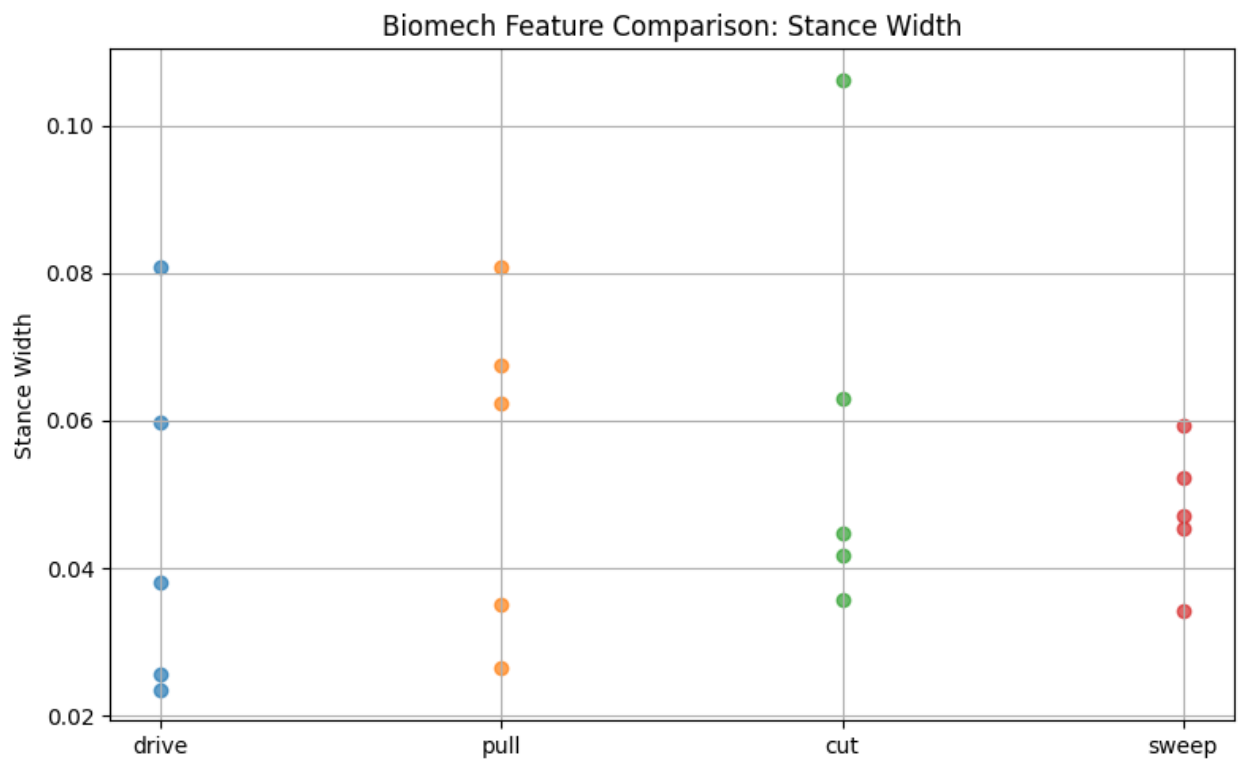
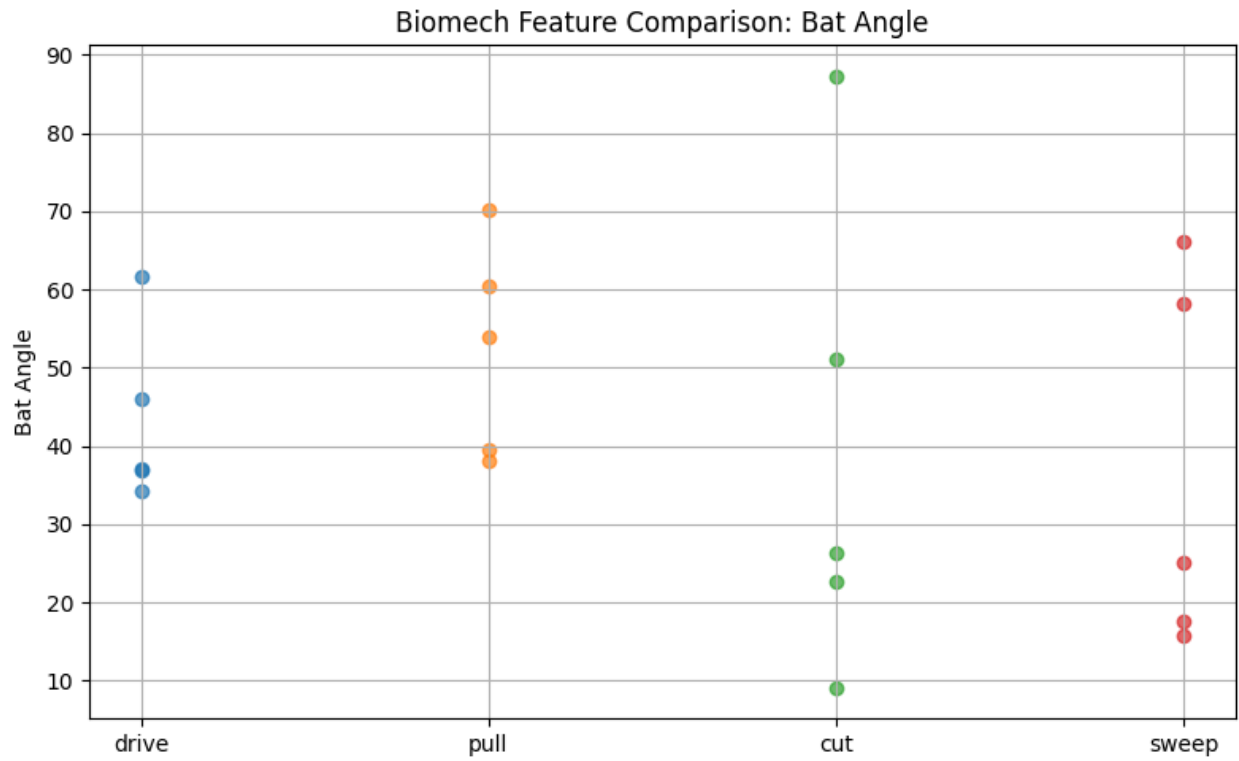
Results Summary

Wrist Velocity

Class	Range	Notes
Drive	0.028–0.054	Moderate motion
Pull	0.009–0.081	Highly variable
Cut	0.023–0.096	Often high
Sweep	0.028–0.073	Moderate to high

 **Observation:** Wrist velocity is noisy and overlaps across all classes. Not a reliable separator alone.





Bat Angle

Class Range Notes

Drive 34–61° Moderate

Pull 38–70° High

Cut 9–87° Very wide spread

Sweep 15–66° Overlaps with Pull

 **Observation:** Bat angle shows some separation (Pull > Drive), but Cut and Sweep overlap heavily. Needs refinement.

Stance Width

Class Range Notes

Drive 0.02–0.08 Narrow

Pull 0.03–0.08 Similar to Drive

Cut 0.03–0.10 Slightly wider

Sweep 0.03–0.06 No clear pattern

 **Observation:** Stance width is inconsistent and likely unreliable due to ankle occlusion or pose noise.

Knee Angle

Class Value Notes

All ~90° Constant across all classes

 **Observation:** Knee angle is not discriminative. Can be dropped.

Strengths

- Biomech features are extracted consistently across clips
- Bat angle and wrist velocity show **some class-specific patterns**

- Diagnostic plots revealed **confusion zones** clearly
 - Model performs well on longer, cleaner clips
-

Limitations

- **Pose ambiguity** in short clips leads to misclassification
 - **Feature overlap** between Drive and Pull is significant
 - **Stance width and knee angle** are weak features
 - **Wrist velocity** is noisy and viewpoint-dependent
 - **Inference confidence** is high even for wrong predictions
-

Next Steps for Improvement

1. Rewrite Feature Logic

- Improve stance width using hip-to-foot or foot-to-foot distance
- Add swing arc or cumulative wrist displacement
- Normalize bat angle using torso orientation

2. Add Confidence Scoring

- Flag clips with <20 valid frames or low wrist motion
- Return “uncertain” if biomech features are ambiguous

3. Train Biomech-Only Classifier

- Use only biomech features to train a shallow model (MLP or SVM)
- Compare its performance against LSTM

4. Retrain with Viewpoint Diversity

- Augment training data with side-on and angled shots
 - Improve generalization across batters and environments
-

Conclusion

This diagnostic audit revealed that **Drive vs Pull confusion is rooted in feature overlap and pose ambiguity**, not just model weakness. By refining biomechanical logic and adding confidence-aware inference, we can significantly improve real-world performance.
